

B.3 Key Continuous Distributions

L1

Continuous Uniform - Basics

$$P(X \in A) = \frac{\text{len of } A}{\text{total len}}$$

$$\text{density} = \frac{1}{\text{len of } S}$$

= If X is uniform from (a, b)

$$f(x) = \frac{1}{b-a}$$

$$F(x) = \frac{x-a}{b-a}$$

$$E[X] = \frac{a+b}{2}$$

$$\text{Var} = \frac{(b-a)^2}{12}$$

remember
this was same
for discrete uniform.

for Var, in discrete,
 $\frac{(\text{no. of values})^2 - 1}{12}$

Eg. $N \sim U(7, 13)$

$$\text{mean} = \frac{7+13}{2} = 10$$

$$\text{Var} = \frac{(13-7)^2}{12} = \frac{6 \times 6}{12} = 3$$

L2: Continuous Uniform - Advance Concepts

- Raw moments can be broken down into pieces. But variance can't be broken down w/o adding the correction term.

$$E[X] = \sum_x x \cdot P(X=x)$$

$$E[X] = \sum_{\text{all } A_i} E[X | X \in A_i] \cdot P(X \in A_i)$$

eg. $X \sim U[0, 100]$. $E[X] = 50$.

$$d = \text{deduct}, \quad Y = \begin{cases} 0 & X \leq d \\ X-d & X > d \end{cases}$$

$$E[X] = 50$$

$$E[Y] = 0.4 \times 50 = 20$$

$$\begin{aligned} P(Y \leq y) &= P(X \leq y + d) \\ &= \frac{y+d}{100} \end{aligned}$$

$$f_Y(y) = \frac{1}{100}$$

$$y > 100$$

$$E[Y] = 0 \cdot P(Y=0) + \int_0^{100-d} y \cdot \frac{1}{100} dy$$

$$= \frac{y^2}{2} \frac{1}{100} \Big|_0^{100-d}$$

$$= \frac{(100-d)^2}{200} = 20$$

$$(100-d)^2 = 4000$$

$$100-d = 20\sqrt{10} \Rightarrow d = 100 - 20\sqrt{10} = 36.75$$

L3 Exponential RV

- X is exponential RV with mean θ if

$$F(x) = 1 - e^{-x/\theta}$$
$$1 - F(x) = e^{-x/\theta}$$

$\lambda = \frac{1}{\theta}$, then

$$F(x) = 1 - e^{-\lambda x}$$

$$f(x) = \frac{d}{dx} F(x) = \lambda e^{-\lambda x} = f(x) \quad x > 0$$

exponentials are used to model waiting times (time btw. hits of a web page, or time btw. rain drops)

$$E[X] = \theta$$

or $\frac{1}{\lambda}$

$$\text{Var}(X) = \theta^2$$

or $\frac{1}{\lambda^2}$

- If X is exponential, then $\text{Var}(X) = (E[X])^2$
- Memoryless Property;

$$P(X > x) = e^{-x/\theta}$$

$$P(X > x+a | X > a) = P(X > x)$$

$$E[X - a | X > a] = E[X] = 0$$

$$\text{Var}[X | X > a] = \text{Var}(X - a | X > a) = \text{Var}(X) = \sigma^2$$

Eg.

Loss $\sim E(50)$

$\theta = 50$

$d = 10$

Median loss given that the payment is made.

$$P(X \leq x | X > 10) = 0.5$$

$$(X - 10 | X > 10) \sim \text{Exp}(50)$$

$$P(X \leq x | X > 10) = 1 - P(X > x | X > 10)$$

$$0.5 = 1 - P(X - 10 > x - 10 | X > 10)$$

$$\ln(0.5) = -\frac{(x-10)}{50}$$

$$x = 44.66$$

L4

Gamma, Exponential & Poisson

$$\text{If } X \sim \text{exp}(\theta)$$

$$f(x) = \frac{1}{\theta} \cdot e^{-x/\theta} \quad x > 0$$

$$\text{If } \boxed{f(x) = c x^{\alpha-1} e^{-x/\theta}}$$

where c should be appropriately chosen to make this a density.

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Now, this becomes a Gamma(α, θ) distribution.

- When α is an integer, it turns out Gamma is the sum of α iid exponentials.

if $\alpha=1$, $\Rightarrow$ exponential
lead term is $x^{\alpha-1}$

$$\alpha=1, \quad f(x) = \frac{1}{\theta} e^{-x/\theta}$$

$$\alpha=2, \quad f(x) = \frac{x}{\theta^2} e^{-x/\theta}$$

$$\alpha=3, \quad f(x) = \frac{1}{2} \frac{x^2}{\theta^3} e^{-x/\theta}$$

Similarly for CDF $\Rightarrow$

$$\alpha=1, \quad F(x) = 1 - e^{-x/\theta}$$

$$\alpha=2, \quad F(x) = 1 - e^{-x/\theta} - \frac{x}{\theta} e^{-x/\theta}$$

$$\alpha=3, \quad F(x) = 1 - e^{-x/\theta} - \frac{x}{\theta} e^{-x/\theta} - \left(\frac{x}{\theta}\right)^2 \frac{1}{2} e^{-x/\theta}$$

$$P(X \leq x) = F(x) = 1 - \sum_{i=0}^{\alpha-1} P(\text{Poisson}(x/\theta) = i)$$

$$\boxed{E[Y] = \alpha \theta}$$

$$\boxed{\text{Var}(Y) = \alpha \theta^2}$$

L5

Beta & Pareto distributions

- X is Beta (a, b)

$$\text{if } f(x) = C x^{a-1} (1-x)^{b-1} \quad 0 < x < 1$$

$$C = \frac{(a+b-1)!}{(a-1)! (b-1)!}$$

$$E[X] = \frac{a}{a+b} \quad E[X^2] = \frac{a(a+1)}{(a+b)(a+b+1)}$$

- Pareto $\rightarrow$ For $\alpha > 0, \theta > 0$

X is Pareto (α, θ)

$$\text{if } f(x) = 0 \quad x < 0 \quad \& \quad \text{for } x > 0$$

$$f(x) = \frac{\alpha \theta^\alpha}{(x+\theta)^{\alpha+1}}$$

$$E[X] = \frac{\theta}{\alpha-1}$$